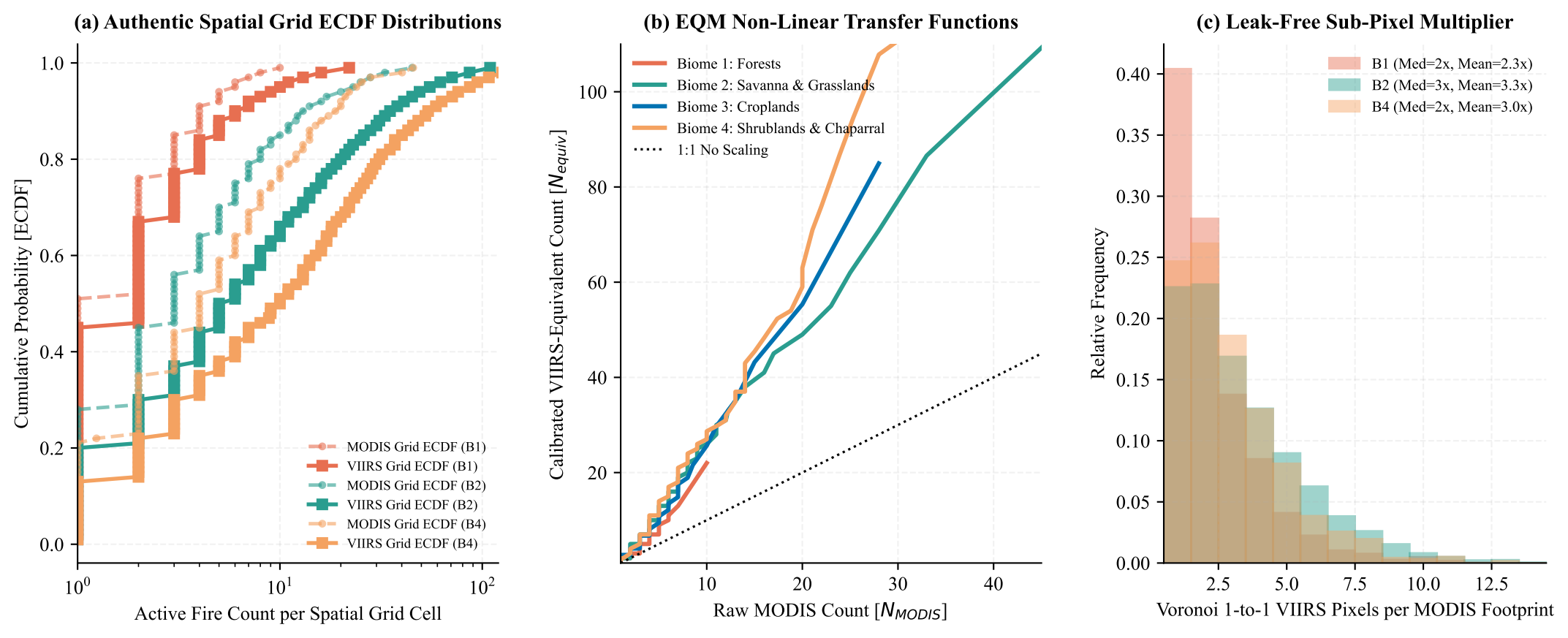

(a) Authentic Spatial Grid ECDF Distributions
Cumulative Probability [ECDF]
Active Fire Count per Spatial Grid Cell
MODIS Grid ECDF (B1)
VIIRS Grid ECDF (B1)
MODIS Grid ECDF (B2)
VIIRS Grid ECDF (B2)
MODIS Grid ECDF (B4)
VIIRS Grid ECDF (B4)
(b) EQM Non-Linear Transfer Functions
Calibrated VIIRS-Equivalent Count [N_{equiv}]
Raw MODIS Count [N_{MODIS}]
Biome 1: Forests
Biome 2: Savanna & Grasslands
Biome 3: Croplands
Biome 4: Shrublands & Chaparral
1:1 No Scaling
(c) Leak-Free Sub-Pixel Multiplier
Relative Frequency
Voronoi 1-to-1 VIIRS Pixels per MODIS Footprint
B1 (Med=2x, Mean=2.3x)
B2 (Med=3x, Mean=3.3x)
B4 (Med=2x, Mean=3.0x)